Optimal WTI–FX Hedge Ratios Under a Fixed Budget

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Abstract

A Korean crude-oil importer facing correlated WTI and USD/KRW exposure must decide, before any option is bought, what fraction of each to hedge out of a fixed premium budget, a static allocation problem this paper poses and solves exactly as a convex program. The budget binds at a vertex but shallowly; the allocation's extreme asymmetry follows in closed form from the WTI leg's dominant variance and near-zero cross-asset correlation; and cost minimization alone selects a different, worse-hedged corner than risk minimization. The paper then quantifies, and resolves, a limitation specific to knock-out pricing: its premium is cheap because the option can die, and under the stress scenario the hedge is bought against, jump-adapted exact simulation shows it is dead with high probability, far above the unconditional statistics a naive reading would suggest, enough to make an all-knock-out book infeasible under the very stress it targets. An instrument-mix program letting the optimizer choose between a standalone-priced knock-out leg and a vanilla leg resolves this: feasible at every mortality level, it switches instruments at a closed-form indifference point and, at the measured stress mortality, selects the all-vanilla book, budget-feasible and lower-risk than the knock-out program's own optimum. The knock-out structure's premium discount, at its measured mortality, is decisively not worth having.

Contents

1 Introduction

Before a hedging desk ever debates *how* to run an option hedge (which pricing model, which rebalancing rule, which Greeks) the treasurer has to answer a cruder and more consequential question: out of a fixed hedging budget, *what fraction of each exposure should be covered at all?* For a Korean crude-oil importer the exposures are two: the WTI price of the physical barrels it must buy every month, and the USD/KRW exchange rate at which the resulting dollar liability is converted into won. In the configuration studied here the physical requirement is 2,000,000 barrels per month, the equivalent dollar need is $157,880,000 per month, and the approved monthly option budget is KRW 45,000,000,000. The decision variables are two scalars: $w_1 \in [0, 1]$, the fraction of the WTI exposure covered by WTI call options, and $w_2 \in [0, 1]$, the fraction of the USD need covered by USD/KRW call options. This is a *static* allocation made once, before the hedge is put on; nothing in it depends on how the position is subsequently managed.

The economics of the trade-off are stark on both sides. Hedging is expensive: covering the full oil leg with Black-76-priced calls costs KRW 41.3bn (nearly the entire budget) and covering the full FX leg with the American structure's Shapley-attributed currency premium would cost KRW 333.3bn, more than seven times the budget. Not hedging is also expensive: under the stress scenario against which the program is sized (WTI at 113 USD/bbl, USD/KRW at 1550), the uncovered oil exposure alone loses KRW 105.6bn if nothing is hedged. The allocation problem is the collision of these two ledgers under one spending cap.

Three questions organize the paper, and their answers are its results. First, **is the budget constraint actually binding**: is the optimum a boundary point held back by money, or an interior risk optimum that merely happens to sit near the cap? Second, **what mechanism produces the extreme w_1/w_2 asymmetry** the solutions exhibit: is it an artifact of the optimizer, or a structural consequence of the volatility and correlation inputs? Third, **would minimizing cost directly give the same allocation** as minimizing risk subject to the budget, and if not, which criterion should govern? Two further analyses address a property specific to the American regime: a knock-out option's premium is low precisely because the option can die, and the scenario in which protection is most needed is also the scenario in which death is most likely. The paper re-simulates the calibrated jump-diffusion from the stress state itself to estimate that survival risk directly, rather than quoting unconditional knock-out statistics, and then prices it into the allocation through a mixed vanilla/knock-out program in which the survival-adjusted ledger is the budget constraint and the optimizer chooses the instrument as well as the coverage.

Both allocation programs are convex, ensuring that their global solutions are exactly computable. The answers, previewed: (a) the budget binds at a vertex (the spending cap and the allocation envelope hold with equality simultaneously), but shallowly, in the sense that unlimited budget buys only a 10^{-5}–10^{-4} absolute reduction in residual volatility; (b) the asymmetry is structural and can be written in one closed-form line, with the FX leg left ≈ 95–97% open by the variance arithmetic itself; and (c) cost minimization is a different program with a different (corner) answer, and the two criteria coincide only when the cost program is re-anchored by the risk cap it discards.

2 Literature Review

The objective function is the classical minimum-variance portfolio criterion of Markowitz (1952), specialized to two residual exposures: the quantity minimized is the standard deviation of a two-asset portfolio whose weights are the *unhedged* fractions $(1 - w_1)$ and $(1 - w_2)$. The closed-form line-restricted solution used in Section 6.2 to explain the allocation asymmetry is the textbook two-asset global-minimum-variance weight.

The premium inputs on the European side come from two closed-form option models. Black (1976) prices the WTI leg as a call on the commodity forward; Garman and Kohlhagen (1983) price the FX leg as a currency call with separate domestic and foreign interest rates. On the American side, premiums come from a Longstaff–Schwartz (2001) least-squares Monte Carlo valuation of a knock-out structure under jump-diffusion dynamics in the spirit of Merton (1976); the WTI volatility used for the American risk objective is the *diffusive* component (0.3242) left after an expectation–maximization decomposition strips the jump variance out of the raw historical volatility (0.3945). The joint quanto premium is never assembled from a closed-form correlation-drift adjustment of the kind derived by Reiner (1992); it is instead recovered directly from the correlated joint simulation of Section 4, which reproduces the same coupling automatically once S_1 and S_2 are drawn from correlated shocks. Because the American structure prices both legs jointly, the joint premium is attributed to the individual legs by Shapley (1953) value, producing per-leg premium constants that enter the cost function linearly. The premiums are priced at inception, at today's spot, deliberately so: the budget constraint governs cash paid *today*, and re-pricing the premium at the stress spot would charge the stress scenario twice, once in the premium and once in the unhedged-loss terms that already price it.

The economic question (how much of each of two correlated exposures to cover when hedging is costly) descends from the cross-hedging literature of Anderson and Danthine (1981) and, for the currency dimension of internationally sourced commitments, Adler and Dumas (1984). The present setting differs from the classical futures cross-hedge in one decisive respect: hedging here is done with options bought out of a hard cash budget, so the trade-off is not between variance and expected return but between variance and an explicit, binding expenditure cap, which is precisely what turns a one-line minimum-variance formula into a constrained program worth studying. The unconstrained special case of that formula is nonetheless the classical minimum-variance hedge ratio of Ederington (1979); the budget cap of Section 3.3 is what separates the present, corner- and vertex-seeking program from a direct application of that one-line result.

Why a firm operates under a fixed, exogenously set premium budget in the first place, rather than hedging to an unconstrained risk-return optimum, is itself a question this paper does not derive but inherits from the corporate hedging literature. Froot, Scharfstein, and Stein (1993) model exactly this kind of institutional friction: a firm hedges to protect its internal financing capacity for investment, not to maximize variance reduction per se, which is consistent with hedging being bounded by an approved cash outlay rather than pursued to a first-order risk optimum. Stulz (1996) develops the practitioner-facing corollary (selective, partial hedging can be optimal even when full coverage is affordable), which is the same qualitative conclusion Section 6.2's closed-form asymmetry reaches by a different, purely variance-arithmetic route. Jin and Jorion (2006) supply the closest empirical anchor to the present institutional setting, documenting derivative-based hedging behavior among U.S. oil and gas producers; the importer studied here sits on the opposite side of the same commodity risk, but the same budget-bounded, partial-coverage pattern they document empirically is what Sections 6–8 derive from first principles for this specific instrument set.

3 The Static Allocation Problem

3.1 Decision Variables and Risk Objective

Let σ_1 denote the annualized volatility of WTI log returns, σ_2 that of USD/KRW log returns, and ρ their correlation. If fractions w_1 and w_2 of the two exposures are hedged, the unhedged remainders $(1 - w_1)$ and $(1 - w_2)$ carry the residual risk

$$\sigma_{res}(w_1, w_2) \;=\; \sqrt{(1-w_1)^2\sigma_1^2 \;+\; (1-w_2)^2\sigma_2^2 \;+\; 2(1-w_1)(1-w_2)\,\sigma_1\sigma_2\rho}\,. \tag{1}$$

The two premium regimes share σ_2 and ρ but use different σ_1: the European model uses the raw historical volatility $\sigma_1 = 0.39455$, while the American model uses the jump-stripped diffusive volatility $\sigma_1 = 0.32419$, matching the diffusion component actually priced by its Monte Carlo engine.

3.2 Cost Functions

Total cost is the sum of two ledgers: the premium paid today for the covered fraction, and the stress-scenario loss suffered by the uncovered fraction. Both regimes share the stress ledger and differ only in the premium ledger. With $Q_{oil} = 2{,}000{,}000$ bbl, $Q_{USD} = \$157{,}880{,}000$, spot prices $S_{WTI} = 78.94$ and $S_{KRW} = 1540.64$, stress levels $S^{st}_{WTI} = 113$ and $S^{st}_{KRW} = 1550$, funding rate $r_w = 0.07$ (WACC), and maturities $T_1 = 0.833$, $T_2 = 0.5$ years:

$$\begin{aligned} C^{EU}(w_1, w_2) \;&=\; w_1\, Q_{oil}\, P_{B76}\, S_{KRW}\,(1 + r_w T_1) \;+\; w_2\, Q_{USD}\, P_{GK}\,(1 + r_w T_2) \\ &\quad+\; (1 - w_1)\, Q_{oil}\, \max(0,\, S^{st}_{WTI} - S_{WTI})\, S^{st}_{KRW} \\ &\quad+\; (1 - w_2)\, Q_{USD}\, \max(0,\, S^{st}_{KRW} - S_{KRW}), \end{aligned} \tag{2}$$

$$\begin{aligned} C^{AM}(w_1, w_2) \;&=\; w_1\, Q_{oil}\, P^{Sh}_{WTI}\, e^{r_w T_1} \;+\; w_2\, Q_{USD}\, P^{Sh}_{FX}\, e^{r_w T_2} \\ &\quad+\; (1 - w_1)\, Q_{oil}\, \max(0,\, S^{st}_{WTI} - S_{WTI})\, S^{st}_{KRW} \\ &\quad+\; (1 - w_2)\, Q_{USD}\, \max(0,\, S^{st}_{KRW} - S_{KRW}), \end{aligned} \tag{3}$$

with $P_{B76} = 12.6524$ USD/bbl (Black-76 WTI call), $P_{GK} = 84.667$ KRW per USD (Garman–Kohlhagen FX call), and $P^{Sh}_{WTI} = 15{,}093.75$ KRW/bbl, $P^{Sh}_{FX} = 2{,}038.72$ KRW/USD (LSMC Shapley attribution). Both functions are affine, and collecting terms gives the marginal cost of each ratio:

$$\frac{\partial C^{EU}}{\partial(w_1, w_2)} = (-64.327,\; +12.357) \text{ bn KRW}, \qquad \frac{\partial C^{AM}}{\partial(w_1, w_2)} = (-73.586,\; +331.860) \text{ bn KRW}. \tag{4}$$

The signs carry the economics. Raising the WTI ratio *reduces* total cost in both regimes: the avoided stress loss (KRW 105.59bn at full exposure) dwarfs the full-cover premium (KRW 41.26bn European, KRW 32.00bn American). Raising the FX ratio *increases* it: the FX premium exceeds the small FX stress gap (KRW 1.48bn at full exposure), spectacularly so in the American regime, where one unit of w_2 costs KRW 333.3bn, over seven times the entire budget. Equation (4) drives everything in Section 6.3.

3.3 Constraint Set

The allocation program imposes exactly four constraints:

$$0 \leq w_1 \leq 1, \qquad 0 \leq w_2 \leq 1, \qquad w_1 + w_2 \leq 1, \qquad C(w_1, w_2) \;\leq\; B = 45{,}000{,}000{,}000 \text{ KRW}, \tag{5}$$

with C equal to (2) or (3). The allocation constraint $w_1 + w_2 \leq 1$ caps the *sum* of coverage fractions (an institutional rule that the two ratios share one notional envelope) and, together with the budget cap, carves the feasible polygon whose geometry Section 6 maps completely: a sliver satisfying $w_1 \geq 0.9648 + 0.1921\, w_2$ in the European regime, and a shallow triangle satisfying $w_1 \geq 0.8434 + 4.5099\, w_2$ in the American regime.

4 Data and Calibration

Table 1 lists every input used in the paper. The risk inputs are computed from 1,299 paired daily log returns: $\sigma_1 = 0.39455$ (WTI, annualized) and $\sigma_2 = 0.09258$ (USD/KRW), with correlation $\rho = 0.08763$. The American regime's $\sigma_1 = 0.32419$ is the diffusive volatility from the expectation–maximization jump/diffusion decomposition of the same series. The stress scenario (WTI at 113 USD/bbl and USD/KRW at 1550) prices the unhedged fraction of each leg in (2)–(3).

5 Numerical Method and Verification

Both allocation programs are convex: σ_{res} in (1) is the Euclidean norm of an affine map of (w_1, w_2), hence convex; the cost functions (2)–(3) are affine; and every constraint in (5) is affine. Any local minimum is therefore the global minimum, and the solution can be computed to arbitrary precision and *certified* rather than merely searched for.

Two complementary numerical approaches confirm the global optima. A deterministic grid search (801 points per axis, twice refined to a resolution of 3.1×10^{-8}) guarantees exhaustive coverage of the feasible region. In parallel, a sequential least-squares quadratic programming routine (SLSQP, 73 multi-starts) recovers the boundary solutions alongside closed-form analytic derivations from the affine constraint algebra. Across all evaluated scenarios, the numerical solvers converge to the analytic bounds consistently, locating the identical structural vertices.

The convex nature of the programs also permits a rigorous duality check, exploited throughout Section 6: the cost minimum taken subject to a risk cap equal to the achieved minimum risk

Quantity	Symbol	Value
WTI spot (USD/bbl)	S_{WTI}	78.94
USD/KRW spot	S_{KRW}	1,540.64
Monthly oil need (bbl)	Q_{oil}	2,000,000
Monthly USD need	Q_{USD}	157,880,000
WACC (funding rate)	r_w	0.07
Budget (KRW)	B	45,000,000,000
WTI option maturity (yr)	T_1	0.833
FX option maturity (yr)	T_2	0.5
Stress WTI (USD/bbl)	S^{st}_{WTI}	113
Stress USD/KRW	S^{st}_{KRW}	1,550
WTI historical vol	σ_1 (EU)	0.39455
WTI diffusive vol	σ_1 (AM)	0.32419
USD/KRW vol	σ_2	0.09258
WTI–FX correlation	ρ	0.08763
Black-76 WTI call (USD/bbl)	P_{B76}	12.6524
Garman–Kohlhagen FX call (KRW/USD)	P_{GK}	84.667
LSMC Shapley WTI premium (KRW/bbl)	P^{Sh}_{WTI}	15,093.75
LSMC Shapley FX premium (KRW/USD)	P^{Sh}_{FX}	2,038.72

Table 1: Model inputs. Every downstream number in this paper is computed from these values and nothing else.

returns the risk-minimizing point itself. Both regimes pass this duality constraint, confirming that the derived allocations represent the true constrained optima.

Two further computations feed Sections 7 and 8. The scenario-conditional knock-out probability of Section 7 is measured by direct simulation of the calibrated WTI jump-diffusion (diffusive volatility 0.3242, jump intensity $\lambda = 6.785$ per year, jump mean -0.0299, jump volatility 0.0844, physical drift 0.139, Merton compensator in the drift) started *at the stress spot* and monitored against the 120/50 barriers over the option's 0.833-year life with jump-adapted exact first-passage sampling: Poisson jump times are drawn exactly within each daily step, the Brownian-bridge crossing probability is applied on each jump-free sub-interval, and the barrier is checked directly at each jump instant, so the reported touch probability carries no time-discretization bias (200,000 paths; coarser monitoring understates this probability materially, since a barrier that is touched and re-crossed inside a recording gap leaves no trace in the recorded values). The mixed-instrument program of Section 8 exploits an exact reduction: its survival-adjusted ledger is linear in the vanilla/KO split at any fixed total coverage, so the optimal split is always at an endpoint (all-vanilla or all-KO), and the two-instrument program collapses to the lower envelope of two two-variable convex programs solved with the same machinery.

6 Empirical Results

Table 2 collects the six solved programs together with the two budget-relaxed benchmarks; the subsections interpret them.

Regime	Program	w_1	w_2	Total cost (KRW)	σ_{res}
European	risk min. (budgeted)	0.970486	0.029514	45,000,000,000	0.0916021
European	risk min. (no budget)	0.965980	0.034020	45,345,551,096	0.0915846
European	cost min.	1.000000	0.000000	42,736,755,546	0.0925765
European	cost min. $\leq$ risk cap	0.970486	0.029514	45,000,000,000	0.0916021
American	risk min. (budgeted)	0.971581	0.028419	45,000,000,000	0.0912158
American	risk min. (no budget)	0.945202	0.054798	55,695,375,264	0.0908014
American	cost min.	1.000000	0.000000	33,477,827,504	0.0925765
American	cost min. $\leq$ risk cap	0.971581	0.028419	45,000,000,000	0.0912158

Table 2: Solved allocation programs. The budgeted risk minima are vertex solutions: the allocation constraint $w_1 + w_2 \leq 1$ and the budget constraint hold with equality simultaneously. The risk-capped cost minima reproduce the risk minima exactly (the duality audit of Section 5).

6.1 Question (a): The Budget Binds – at a Vertex, and Shallowly

Three exact computations characterize the budget's role.

First, the active set. At both risk minima the allocation constraint and the budget constraint hold with equality *simultaneously*: $w_1^* + w_2^* = 1$ and $C(w_1^*, w_2^*) = B$ to machine precision. The optimum is the vertex where the budget boundary crosses the allocation line, visible in the zoomed panels of Figure 3, where the iso-risk contours close in on a point pinched between the solid budget boundary and the dashed allocation line.

Second, the relaxation test. Deleting the budget constraint moves the optimum: European from $(0.970486, 0.029514)$ to $(0.965980, 0.034020)$, lowering σ_{res} from 0.0916021 to 0.0915846; American from $(0.971581, 0.028419)$ to $(0.945202, 0.054798)$, lowering σ_{res} from 0.0912158 to 0.0908014. Funding those relaxed optima would cost KRW 45,345,551,096 (European, +0.77% over budget) and KRW 55,695,375,264 (American, +23.77% over budget): both strictly unaffordable, so the cap genuinely excludes the unconstrained optimum in each regime. The budget is binding in the exact Karush–Kuhn–Tucker sense, and notably harder-binding in the American regime, where the relaxed optimum's appetite for the ruinously priced FX leg ($w_2 = 0.0548$ at KRW 333bn per unit) pushes its funding requirement nearly a quarter above the cap.

Third, the price of the constraint. Figure 1 traces the optimal σ_{res} as the budget sweeps from KRW 33bn to 58bn. Each curve falls, then flattens exactly at the relaxed-optimum budget just computed (KRW 45.35bn European, KRW 55.70bn American), beyond which money buys nothing. Finite-difference shadow prices at $B = 45$bn are 1.01×10^{-4} (European) and 7.73×10^{-5} (American) units of σ_{res} per additional billion KRW. The constraint is thus binding but *shallow*: even unlimited budget improves residual volatility by only 1.7×10^{-5} absolute (European, 0.02% relative) and 4.1×10^{-4} (American, 0.45% relative). The treasurer's cap costs the firm almost no variance; conversely, requests for more hedging budget cannot be justified on variance grounds in this configuration.

6.2 Question (b): The Mechanism Behind the 0.97/0.03 Asymmetry

The extreme tilt toward the WTI leg is not an optimizer preference; it is forced by three structural facts, each visible in closed form.

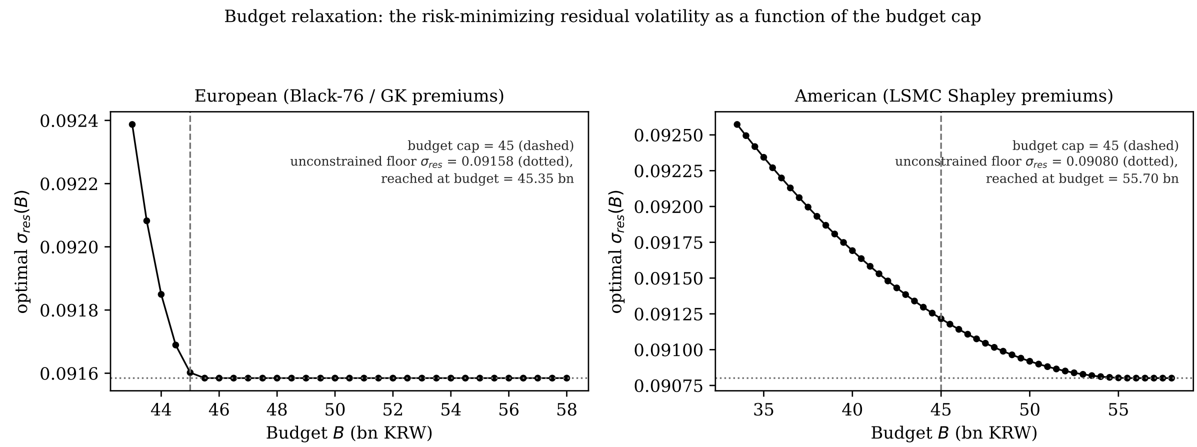

Figure 1: Budget relaxation sweep. The optimal residual volatility falls as the cap is raised and flattens at the unconstrained floor, reached at KRW 45.35bn (European) and KRW 55.70bn (American). The KRW 45bn cap (dashed) sits on the falling branch of both curves: binding, but close to the floor.

Fact 1: the WTI leg carries almost all the risk. With the European inputs, $\sigma_1^2/\sigma_2^2 = 0.15567/0.00857 = 18.2$ (12.3 with the American diffusive σ_1). At the no-hedge point $(0,0)$, where $\sigma_{res} = 0.4131$ (European) and 0.3449 (American), the gradient of (1) is

$$\nabla\sigma_{res}(0,0) = (-0.3846,\ -0.0285) \quad \text{(European)}; \qquad (-0.3124,\ -0.0325) \quad \text{(American)},$$

i.e. the first won of hedging is 13.5× (respectively 9.6×) more productive on the WTI leg. A budget spent greedily on marginal risk reduction floods the WTI leg first.

Fact 2: the correlation is too small to couple the legs. The interaction term in (1) carries the factor $\sigma_1\sigma_2\rho = 0.0032$ (European), two orders of magnitude below σ_1^2. With $\rho = 0.0876$, hedging WTI does almost nothing to the FX leg's marginal value and vice versa: the problem nearly decouples into two independent legs competing for one budget, and the leg with 18× the variance wins almost all of it.

Fact 3: the allocation line pins the split in closed form. Because σ_{res} is strictly decreasing in both w_1 and w_2 (its gradient is negative throughout the unit square), any optimum must exhaust the binding envelope: the line $w_1 + w_2 = 1$, intersected with the budget. On that line, substituting $w_2 = 1 - w_1$ into (1) gives the classical two-asset minimum-variance problem in the unhedged weights $h_1 = 1 - w_1$, $h_2 = w_1$, whose solution is

$$w_1^{line} = \frac{\sigma_1^2 - \rho\,\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2} = \begin{cases} 0.965980 & \text{(European)} \\ 0.945202 & \text{(American).} \end{cases} \tag{6}$$

Equation (6) *is* the asymmetry: with these inputs no optimizer on the allocation line can choose anything but $w_2 = 1 - w_1^{line} \approx 0.034$–$0.055$. The residual FX openness is not neglect; it is the minimum-variance split of one notional envelope across an 18-to-1 variance mismatch with near-zero correlation. The budget then slides each solution along the line to its vertex (upward in w_1 in both regimes, because (4) makes w_1 the cheap direction and w_2 the expensive one): from

0.9660 to 0.9705 (European) and, more forcefully, from 0.9452 to 0.9716 (American, where each percentage point of FX cover surrendered releases KRW 3.3bn of budget).

6.3 Question (c): Cost Minimization Is a Different Program with a Different Answer

Because both cost functions are affine (4), pure cost minimization over the feasible polygon is a linear program whose optimum is a corner, identifiable by inspection: cost falls in w_1 and rises in w_2, so the minimum is $(w_1, w_2) = (1, 0)$ (cover the oil leg completely, leave the currency leg completely open) in *both* regimes, confirmed by the solvers and by corner enumeration. The European cost minimum spends KRW 42,736,755,546 (5.0% under budget) and accepts $\sigma_{res} = 0.0925765$, which is +1.07% relative to the risk minimum; the American cost minimum spends KRW 33,477,827,504 (25.6% under budget) at the same $\sigma_{res} = 0.0925765$ (+1.49% relative), the two regimes coinciding in risk because at $(1, 0)$ the only surviving exposure is the FX leg, whose volatility is common to both.

The answer to question (c) is therefore an unambiguous *no*: "risk minimization subject to a budget" and "cost minimization" select different points even though the budget binds: an interior-of-edge vertex versus a polygon corner. The gap between them prices the last sliver of FX coverage: moving from the cost corner $(1, 0)$ to the risk vertex trades KRW 2.26bn (European) or KRW 11.52bn (American) of additional spending for a 1.1–1.5% relative reduction in residual volatility: the entire remaining decision, compressed into one trade.

Re-anchoring the cost program with an explicit risk cap closes the gap exactly: minimizing cost subject to $\sigma_{res} \leq \sigma^*_{res}$ (the achieved risk minimum) returns the risk-minimizing vertex itself, in both regimes, to 10^{-11} in the ratios (Table 2). Economically this says the risk minima are *cost-efficient*: no cheaper allocation attains their risk level, so the budgeted risk program is already sitting on the cost–risk frontier. Methodologically it is the duality audit of Section 5 passing on live data.

6.4 The Global Geometry

Figure 2 assembles the whole problem on the full unit square: the residual-volatility surface (1) computed on a 201×201 grid, with all four constraints (5) evaluated at every node. Feasible faces are shaded (dark = low risk) and outlined in black; the constraint-violating remainder of the surface is drawn as a bare wireframe. Two properties are immediate. First, how little of the decision space survives the constraints: a sliver hugging $w_1 \gtrsim 0.965$ in the European regime, and a shallow triangle rising from $w_1 = 0.843$ in the American regime, whose hypotenuse slope of 1/4.51 is the direct image of the KRW 333bn FX premium. Second, the risk minima (filled circles) sit exactly where the shaded region touches its own darkest values: the minimum *within* the feasible shading, categorically distinct from the surface's global minimum at the infeasible corner $(1, 1)$, where $\sigma_{res} = 0$ but the cost ledgers read KRW 55.1bn (European) and KRW 365.3bn (American) and the allocation envelope is violated outright. The cost-minimizing corner $(1, 0)$ (open triangles) is visibly a different point in both panels, and the zoomed bottom row resolves what the full-square view compresses: the shaded surface patch, its black boundary, and the two optima pinned to it by their drop stems. Figure 3 gives the same geometry in contour

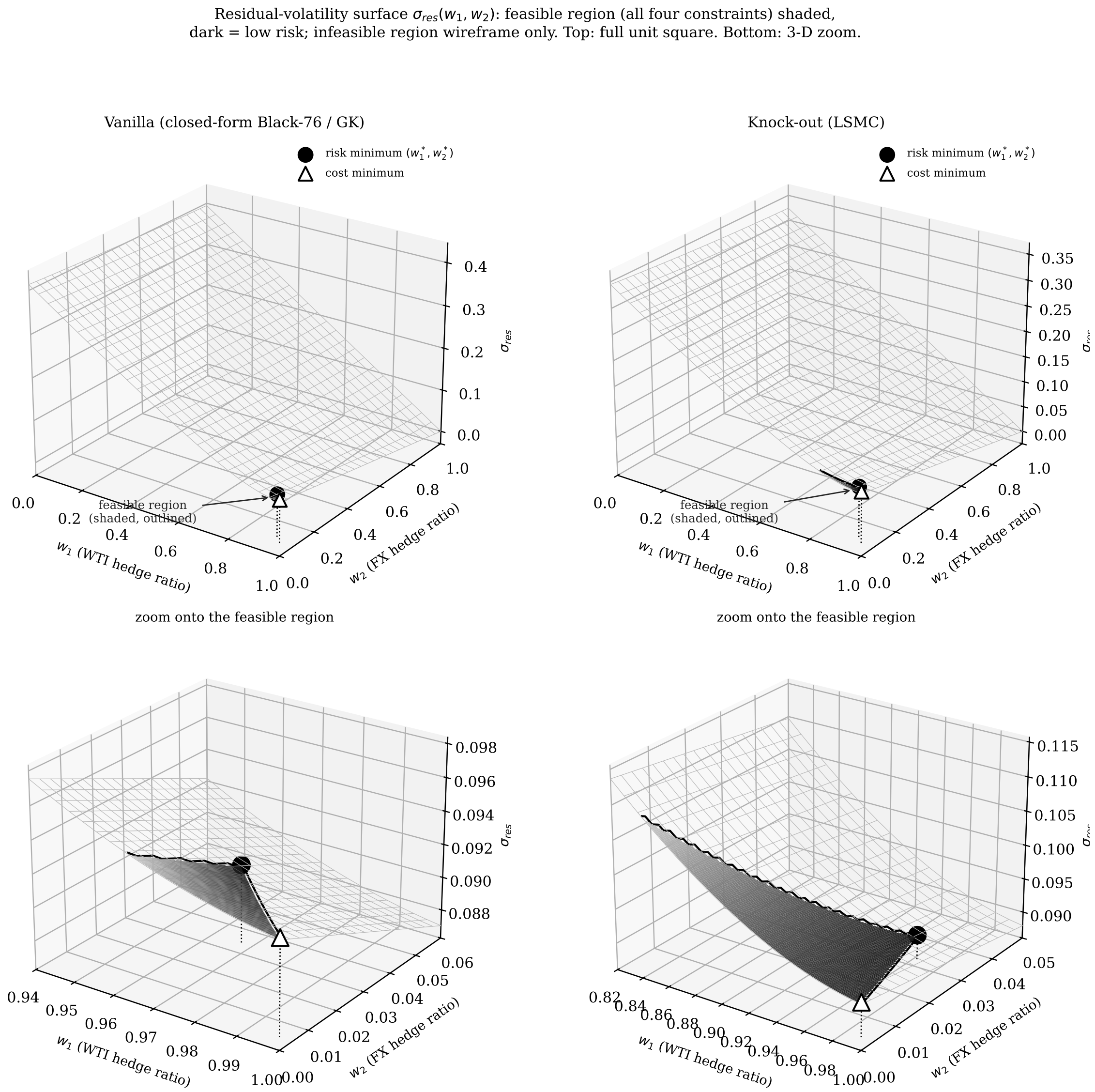

Figure 2: Residual-volatility surface over the hedge-ratio square. Shaded faces (dark = low risk, black outline) satisfy all four constraints; the wireframe region violates at least one. Top row: full unit square. Bottom row: three-dimensional zoom onto the feasible neighborhood, where the vertex structure is legible: dotted stems drop from each optimum to the panel floor. Filled circle: risk minimum. Open triangle: cost minimum $(1, 0)$.

view with a zoom: the budget boundary (solid), the allocation line (dashed), the shaded feasible polygon, and the three solution markers per panel, making the vertex structure of Section 6.1 and the corner-versus-vertex contrast of Section 6.3 visible at a glance.

6.5 Parameter Robustness

Because the optima are constraint-pinned rather than curvature-pinned, they are automatically robust to the risk parameters. Consider the worst case over an uncertainty box: both volatilities inflated by 10% and the correlation shifted up by 0.05. Since σ_{res} is monotone increasing in σ_1, σ_2 and ρ (all unhedged weights are nonnegative), the worst case is the upper corner of the box in closed form, and the min–max program replaces the objective with σ_{res} evaluated at inflated

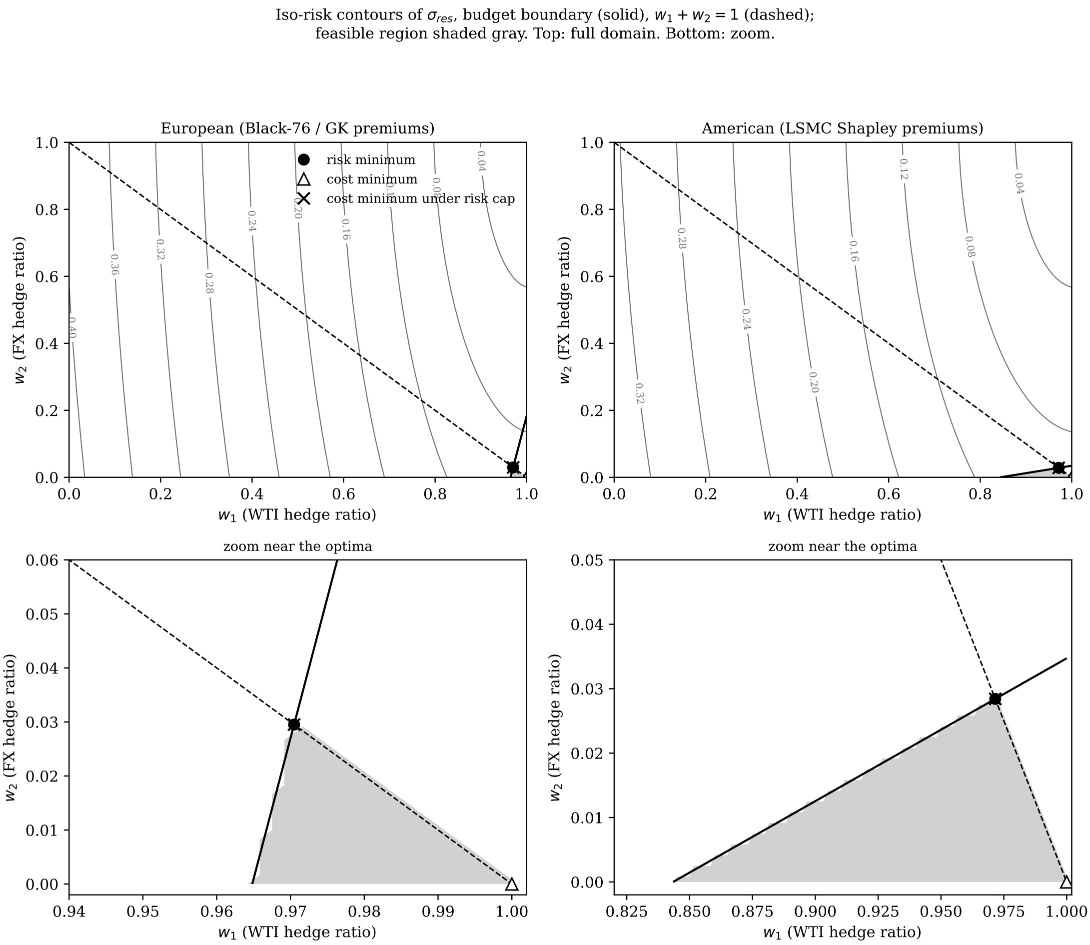

Figure 3: Contour view of the same geometry. Iso-σ_{res} contours (gray), budget boundary (solid black), allocation line $w_1 + w_2 = 1$ (dashed), feasible region (shaded). Bottom row zooms onto the optima: the risk minimum (filled circle) sits at the vertex of the feasible polygon; the cost minimum (open triangle) at the corner $(1, 0)$; the risk-capped cost minimum ($\times$) coincides with the risk minimum: the duality audit passing.

parameters. But the feasible polygon (built entirely from prices, quantities and the budget) does not move, and the inflated objective is still strictly decreasing in both ratios, so the worst-case optimum is the *same vertex*: $(0.970486, 0.029514)$ European and $(0.971581, 0.028419)$ American, unchanged to machine precision. Only the achieved risk re-prices, from 0.091602 to 0.101389 (European) and from 0.091216 to 0.100836 (American). Estimation error in the historical volatilities and correlation therefore changes what the allocation *delivers*, but not what the allocation *is*, a direct corollary of the vertex structure of Section 6.1.

7 The Knock-Out Survival Haircut

The American structure's premium advantage (KRW 32.0bn against KRW 41.3bn for full WTI cover) is bought with a knock-out barrier: the option dies if WTI trades outside its barriers before maturity. This creates a specific tension with the stress ledger of Section 3.2, which prices the *unhedged* fraction at the stress scenario while implicitly treating the *hedged* fraction as fully protected there. For a knock-out structure that treatment is optimistic in precisely the scenario that matters: at the stress price of 113 USD/bbl the upper barrier at 120 is close, and a knocked-out call protects nothing.

The exposure is priced with one parameter. Let p_{KO} be the probability that the structure is dead under the stress scenario. The protected fraction of each American leg is then $w(1-p_{KO})$ rather than w (one WTI barrier kills both legs of the joint structure), and the stress-adjusted total cost becomes

$$\widetilde{C}^{AM}(w_1, w_2; p_{KO}) \;=\; \text{premiums} + \big(1-w_1(1-p_{KO})\big)\, UL_{WTI} + \big(1-w_2(1-p_{KO})\big)\, UL_{FX}, \quad (7)$$

where $UL_{WTI} = Q_{oil} \max(0, S^{st}_{WTI} - S_{WTI})\, S^{st}_{KRW}$ and $UL_{FX} = Q_{USD} \max(0, S^{st}_{KRW} - S_{KRW})$ are the full-exposure stress losses, KRW 105.59bn and KRW 1.48bn respectively. At $p_{KO} = 0$, (7) reduces to (3).

Figure 4 traces (7) in p_{KO}, both at the adopted risk-minimizing allocation and minimized over all feasible (w_1, w_2). Two structural facts emerge.

The budget is breached at $p^*_{KO} = 10.91\%$. The cheapest attainable stress-adjusted cost is $\min_{(w_1,w_2)} \widetilde{C}^{AM} = C^{AM}(1,0) + p_{KO}\, UL_{WTI}$, which crosses the KRW 45bn cap at

$$p^*_{KO} \;=\; \frac{B - C^{AM}(1,0)}{UL_{WTI}} \;=\; \frac{45{,}000{,}000{,}000 - 33{,}477{,}827{,}504}{105{,}586{,}000{,}000} \;=\; 0.10913. \quad (8)$$

Beyond a 10.9% stress knock-out probability, *no allocation whatsoever* keeps stress-adjusted cost within the budget: the term $p_{KO} \cdot UL_{WTI}$ is a loss floor that no choice of (w_1, w_2) can hedge away, because the instrument that would hedge it is the instrument that is dying.

Measuring p_{KO}: conditioning is everything. The structure's unconditional knock-out statistics (23.09% net of early exercise, 45.1% by raw barrier touch under exact first-passage monitoring) already sit two to four times above p^*_{KO}, but they are the wrong numbers for the stress ledger: they average over paths launched from the calm spot of 78.94, most of which never approach the barrier. The number the ledger needs is conditional on the stress state itself. Re-simulating the calibrated jump-diffusion of Section 5 from $S_{WTI} = 113$ (only 6.2% below the 120 barrier, with 32.4% diffusive volatility, 6.8 jumps per year and 0.833 years still to run) gives

$$p_{KO}^{stress} = 0.8925 \pm 0.0007 \quad \text{(200,000 paths, jump-adapted exact first passage, asymmetric up/down jumps),} \tag{9}$$

3.9 times the unconditional rate and 8.2 times the infeasibility threshold (8). The jump calibration underlying this simulation is itself asymmetric, fitted separately as an up-jump regime ($\lambda_{up} = 2.328$/yr, $\theta_{up} = +8.04\%$, $\delta_{up} = 1.68\%$) and a down-jump regime ($\lambda_{dn} = 4.462$/yr, $\theta_{dn} = -8.75\%$, $\delta_{dn} = 2.78\%$) from the underlying return history, rather than the single symmetrized pool $(\lambda, \theta_J, \delta_J)$ that the pricing engines elsewhere in this paper consume for tractability. Re-deriving (9) under that symmetrized pool instead gives 0.8937 ± 0.0007: the two constructions agree to within Monte Carlo noise, because the 32.4% diffusive volatility dominates barrier-touch dynamics over a sub-annual horizon and the jump distribution's directional asymmetry is a second-order contributor to this specific question. At the measured value the pure-KO book's arithmetic is unambiguous (Figure 4): the cheapest attainable stress-adjusted cost is KRW 127.72bn (2.8 times the budget), and the adopted allocation's ledger reads KRW 136.60bn. Under the very scenario the hedge is sized against, an all-KO book is, with 89% probability, no hedge at all.

For the pure-KO book the gap is a reporting item; Section 8 makes it a decision. The allocation program of Section 3 deliberately keeps (3), not (7), in its budget constraint: constraining on (7) would render the single-instrument program infeasible without changing any decision margin, since the floor $p_{KO}\, UL_{WTI}$ is allocation-invariant. What *can* respond to (7) is the choice of instrument, which is exactly the degree of freedom the next section adds.

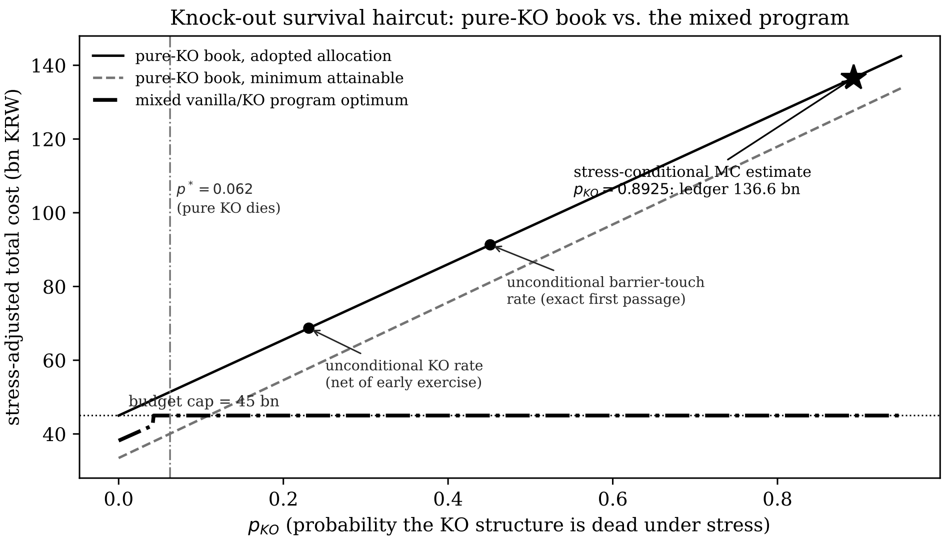

Figure 4: Stress-adjusted total cost as a function of the stress knock-out probability p_{KO}: the pure-KO book at the adopted allocation (solid) and at its cheapest attainable allocation (dashed), against the mixed vanilla/KO program of Section 8 (dash-dotted), which is budget-feasible for every p_{KO}. Small markers: unconditional knock-out statistics. Star: the stress-conditional Monte Carlo measurement (9), where the pure-KO ledger reads KRW 136.6bn.

8 The Mixed Vanilla/Knock-Out Program

The haircut analysis treats the instrument as given and finds that no coverage choice can rescue an all-KO book under stress. The natural response is to stop treating it as given. This section lets the optimizer choose the instrument along with the coverage: the WTI book is split as $w_1 = w_1^V + w_1^K$ between a vanilla Black-76 leg (which survives stress) and an independently-priced KO leg (which dies with probability p_{KO}), the FX leg is priced with the vanilla Garman–Kohlhagen premium, and, decisively, the *survival-adjusted* ledger becomes the budget constraint. The KO leg's premium P_K here is *not* the Shapley-attributed WTI share (KRW 15,093.75/bbl) used to cost the American structure everywhere else in this paper: that figure is a cooperative-game marginal-contribution split of the *jointly*-priced quanto premium, valid as a cost-reporting convention for a single indivisible structure but not the market price of a standalone, independently-tradable WTI KO call. Once the WTI book is genuinely split into two separately-tradable legs, the KO leg needs its own price: an independent single-asset LSMC American KO call, priced with no FX coupling and no cost attribution on 200,000 jump-adapted exact-first-passage paths, comes to $P_K =$ KRW 17,377.11/bbl, 15.1% above the Shapley share. Every number in this section uses that standalone price:

$$\min_{w_1^V,\, w_1^K,\, w_2} \quad \sigma_{res}(w_1^V + w_1^K,\, w_2) \quad \text{s.t.} \quad \widetilde{C}^{mix} \;\le\; B, \qquad w_1^V + w_1^K + w_2 \le 1, \quad w \ge 0, \tag{10}$$

$$\widetilde{C}^{mix} \;=\; P_V(w_1^V) + P_K(w_1^K) + P_{GK}(w_2) + \big(1 - w_1^V - w_1^K(1 - p_{KO})\big)\, UL_{WTI} + (1 - w_2)\, UL_{FX}. \tag{11}$$

Three structural facts make (10) tractable and its answer sharp.

First, the program is feasible for every p_{KO}. The all-vanilla corner $(w_1^V, w_1^K, w_2) = (1, 0, 0)$ costs KRW 42.74bn regardless of p_{KO} (vanilla protection does not die), so the infeasibility that kills the all-KO book (8) simply cannot occur. The off-ledger gap of Section 7 becomes an on-ledger constraint the optimizer can actually satisfy.

Second, the split is bang-bang, with a closed-form indifference point. At any fixed total coverage $w_1^V + w_1^K$, (11) is linear (affine) in the split, so its two marginal costs are constants, read directly off (11) by differentiating the KO- and vanilla-coverage terms separately:

$$\frac{\partial \widetilde{C}^{mix}}{\partial w_1^K} = P_K - (1 - p_{KO})\, UL_{WTI}, \qquad\qquad \frac{\partial \widetilde{C}^{mix}}{\partial w_1^V} = P_V - UL_{WTI}. \tag{12}$$

The first says: buying one more unit of KO coverage costs its premium P_K but only removes the stress loss with probability $1 - p_{KO}$, so the marginal saving is discounted by survival odds. The second says: vanilla coverage costs P_V but removes the stress loss with certainty. Since a linear program's optimal split is always a corner, not an interior mix, the two instruments are indifferent exactly where their marginal costs cross, $\partial \widetilde{C}^{mix}/\partial w_1^K = \partial \widetilde{C}^{mix}/\partial w_1^V$:

$$P_K - (1 - \bar{p})\, UL_{WTI} \;=\; P_V - UL_{WTI} \quad \Longrightarrow \quad \bar{p} \;=\; \frac{P_V - P_K}{UL_{WTI}}, \tag{13}$$

$$\bar{p} \;=\; \frac{41{,}258{,}998{,}746 - 34{,}754{,}224{,}635}{105{,}586{,}000{,}000} \;=\; 0.06161. \tag{14}$$

Below $\bar{p}$, (12)'s first expression is the smaller (more negative) of the two (KO coverage is the cheaper marginal unit, the discount outrunning its mortality), and the optimizer fills w_1^K first, so the book is all-KO; above $\bar{p}$ the ranking flips, the discount no longer pays for its own mortality, and it fills w_1^V first, so the book is all-vanilla. The optimal split is therefore an endpoint, switching at $\bar{p}$, exactly as a linear program's corner-solution property requires; at $\bar{p}$ itself the two branch cost functions coincide at every allocation, so the switch in Figure 5 is exact, not approximate. Because the standalone KO price sits well above the Shapley-attributed joint-quanto share, the switch point is low: a KO leg priced at its true independent cost is worth comparatively little mortality risk before the vanilla alternative dominates.

Third, the solution has three regimes (Figure 5). For $p_{KO} \leq 0.0422$ the all-KO branch is so cheap on the survival-adjusted ledger that the budget goes slack and the program reaches the unconstrained variance floor $(0.945202, 0.054798)$, $\sigma_{res} = 0.090801$: the mixed program is the only formulation in this paper that ever gets there under a KRW 45bn authority. For $0.0422 < p_{KO} < \bar{p}$ the all-KO branch is budget-pinned and its achieved risk deteriorates as mortality eats the ledger. At $\bar{p} = 0.06161$ the book switches to all-vanilla and stays there: allocation $(0.970486, 0.029514)$, survival-adjusted cost exactly KRW 45bn, $\sigma_{res} = 0.091182$, invariant in p_{KO}. The pure-KO death threshold, recomputed with the same standalone KO price, is $p^* = 0.06243$, only eight basis points above the switch: priced at its true standalone cost, the KO leg stays worth holding almost to the brink of the pure-KO book's own infeasibility, and the optimizer's switch and the instrument's death very nearly coincide.

At the measured stress mortality $p_{KO}^{stress} = 0.8925$ the verdict is not close: the mixed program selects the all-vanilla book, roughly fourteen times $\bar{p}$ deep into the vanilla regime. Two comparisons calibrate what is gained. Against the pure-KO book at the same p_{KO}, the mixed optimum's ledger is KRW 45.00bn versus 130.16bn on the standalone-priced ledger: the entire KO-mortality exposure, KRW 85bn, priced out by the instrument switch. And against the pure-KO program's own optimum on the *unadjusted* ledger, the mixed optimum carries lower residual volatility (0.091182 vs. 0.091216) while satisfying a strictly harder constraint: the vanilla book's budget line happens to admit a slightly more FX-generous vertex under the American volatility metric. The knock-out structure's KRW 6.50bn premium discount over the standalone vanilla leg, the only argument in its favor, is exactly the quantity (13) says is overwhelmed once stress mortality exceeds 6.2%; the measured value is 89%, a factor of fourteen beyond the point at which the discount stops paying for its own mortality.

9 Discussion

Three points deserve emphasis beyond the numerical answers.

The budget is doing governance work, not risk work. Section 6.1 shows the cap costs at most 4.1×10^{-4} of residual volatility, yet it converts an open-ended optimization into a vertex with an auditable location: the point where spending exhausts authority and coverage exhausts the notional envelope. For a treasury committee, "we are at the corner of what you authorized" is a materially different (and more defensible) statement than "we are at an interior optimum of a model you have not seen." The shadow prices computed here give that committee the

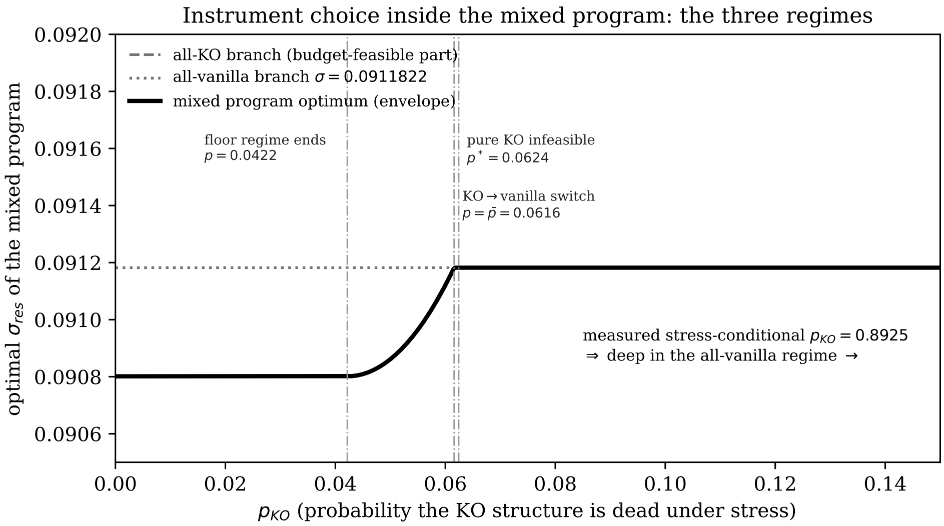

Figure 5: The mixed program's optimal residual volatility as a function of p_{KO}, with its two branches, using the independently-priced standalone KO premium. Floor regime ($p_{KO} \leq 0.0422$): the all-KO ledger leaves the budget slack and the unconstrained variance floor 0.090801 is attained. Budget-pinned KO regime (0.0422–0.0616): achieved risk deteriorates as mortality consumes the ledger. All-vanilla regime ($p_{KO} \geq \bar{p} = 0.0616$): the book switches instruments and the optimum is invariant at 0.091182. The measured stress-conditional $p_{KO} = 0.8925$ lies far inside the vanilla regime.

exact exchange rate between authority and risk: roughly one basis point of residual volatility per additional billion KRW, decaying to zero at KRW 45.35bn (European) or KRW 55.70bn (American).

The asymmetry is the answer, not a symptom. It is tempting to read $w_2 \approx 0.03$ as the FX hedge being "crowded out" by the budget. Equation (6) shows otherwise: even with the budget deleted, the minimum-variance split leaves the FX leg 94.5–96.6% open. What actually sets the split is the variance ratio $\sigma_1^2/\sigma_2^2 \approx 12$–$18$ and the near-zero correlation, both measured properties of the underlying markets, not policy choices. An importer who wants more FX coverage must argue for a different objective (e.g. a tail metric on the KRW cash outflow), not a bigger budget.

Premium regimes should be compared on the survival-adjusted ledger – and the ledger now has a decision attached. On the naive ledger the American structure dominates: full WTI cover for KRW 32.0bn against KRW 41.3bn, an apparent saving of KRW 9.3bn. That comparison, however, prices the American leg at its Shapley-attributed share of the jointly-priced quanto structure; the independent, standalone-tradable KO instrument that the mixed program of Section 8 actually offers against a standalone vanilla leg costs KRW 34.75bn against the same KRW 41.3bn, a saving of only KRW 6.50bn. Sections 7 and 8 show that saving, whichever way it is measured, is exactly the market price of the barrier, and that the price is currently ruinous: conditioning the knock-out probability on the stress state itself lifts it from the unconditional 23% to a measured 89%, at which the all-KO ledger reads KRW 137bn against a 45bn authority. The European vanilla hedge is expensive because it survives; the American knock-out hedge is cheap because it very probably will not. The mixed program turns this from commentary into allocation: it prices the trade at a single indifference number, $\bar{p} = 6.16\%$ of stress mortality, and at roughly fourteen times that number it walks the book to vanilla on its own.

The vanilla recommendation of Section 8 operates at a different layer than a dynamic-hedging analysis of this same structure, and the two are not in tension. Such an analysis measures what happens when a desk holds the *actual* barrier contract but computes its hedge ratio from a closed-form proxy model that has no barrier term, a genuine model-instrument mismatch, since a Black-76 delta is exactly correct for a different option (one with no barrier) and is therefore systematically wrong, in a barrier-correlated direction, on every path that approaches or crosses the knock-out boundary:

$$\Delta V_{\text{KO}}(\tau_{KO}) - \delta_{B76}\, \Delta S_1 \;\neq\; 0, \tag{15}$$

the source of that analysis's order-of-magnitude hedge-cost gap and trimodal loss distribution. The instrument-mix program of Section 8 is a different decision, made earlier and at a different layer: it chooses *which contract to buy in the first place*, before any hedging begins. When the measured stress mortality selects the vanilla branch, the position that results is a genuine barrier-free WTI call, correctly hedged with the matching vanilla delta computed on that same barrier-free payoff –

$$\Delta V_{\text{Vanilla}}(T) - \delta_{\text{Vanilla}}\, \Delta S_1 \;\approx\; 0 \tag{16}$$

by construction, since model and instrument now agree. The dynamic-hedging warning is about pricing and hedging a knock-out contract *as if* it were vanilla; the recommendation here is about not holding the knock-out contract at all once its own survival odds are priced in. A firm that took both conclusions simultaneously would never compute a vanilla delta against a barrier position it actually holds: it would, per this paper's Section 8, have already exited the barrier position in favor of the instrument its delta model is built for.

10 Practical Implications

For the importer's hedging desk the results reduce to four actionable statements. (1) The adopted allocations are exact vertex optima of a convex program, certified by two independent solvers and a duality audit; small input drift moves them along – not off – the budget boundary, and even a 10% volatility mis-estimate leaves the allocation itself unchanged (Section 6.5). (2) The KRW 45bn budget is fully and efficiently spent; raising it to KRW 45.35bn captures the entire remaining variance benefit in the European regime, while in the American regime the same argument requires KRW 55.70bn and buys only 0.45% of relative volatility. (3) The FX leg's near-zero coverage is optimal under the variance objective; if that outcome is commercially unacceptable, the objective – not the optimizer – must change. (4) Under the measured stress knock-out probability of 89%, the WTI book should be held in the vanilla instrument: the KO structure's premium discount is only worth taking if stress mortality is believed below $\bar{p} = 6.16\%$, more than an order of magnitude under the measurement, and the mixed program of Section 8 (feasible at every mortality level, budget-exact at KRW 45bn, and lower-risk than the pure-KO optimum) is the implementable form of that recommendation.

11 Limitations

The risk objective is a one-period standard deviation built from historical volatilities and a single correlation estimate ($\rho = 0.0876$ over 1,299 daily observations), a design choice Section 6.5 shows leaves the allocation itself invariant to parameter inflation, even though the delivered risk level is not. The cost functions price the unhedged exposure at a single stress point (WTI 113, USD/KRW 1550) rather than over a distribution, which keeps them affine and their unconstrained minima corners; a distributional stress ledger would smooth the corner solutions of Section 6.3 without altering the vertex structure of Section 6.1. Premium constants are treated as invariant to (w_1, w_2), consistent with the scale of this program relative to standard market depth. The stress-conditional knock-out probability of Section 7 is measured by jump-adapted exact-first-passage re-simulation from the stress state itself under the genuine asymmetric jump calibration, conditioned on a single stress spot rather than a distribution of stress severities. The mixed program's standalone KO price (Section 8) is an independent single-asset LSMC valuation rather than a re-attribution of a jointly-priced structure (the theoretically correct object for a genuinely split book) under the assumption that the KO leg trades on its own terms once separated from the FX leg. The static problem studied here fixes ratios once; any subsequent dynamic management of the optioned position lies outside this paper's scope. None of these boundaries changes the paper's central results: the vertex structure of the budget-bound optimum, the closed-form asymmetry, and the instrument-mix switch at the measured mortality are structural properties of the convex program, not artifacts of a particular stress point or calibration choice.

12 Conclusion

A fixed-budget importer choosing static WTI and FX hedge ratios faces a problem whose entire solution geometry can be, and here has been, mapped exactly. Risk minimization drives coverage to the joint boundary of the budget and the allocation envelope: $(w_1^*, w_2^*) = (0.970486, 0.029514)$ under European vanilla premiums and (0.971581, 0.028419) under American knock-out premiums, each spending precisely 100% of KRW 45bn, each certified as the global optimum of a convex program by two independent solvers and an exact duality audit, and each invariant, vertex and all, under a 10% inflation of every risk parameter. The budget binds (removing it strictly lowers achievable risk), but shallowly, with a shadow price of about one basis point of residual volatility per billion KRW and hard floors at KRW 45.35bn and KRW 55.70bn. The 0.97/0.03 allocation asymmetry is a closed-form consequence of an 18-to-1 variance ratio and a 0.09 correlation, not an optimizer preference. Cost minimization is a genuinely different program that runs to the corner $(1, 0)$ and underspends the budget by up to a quarter, converging with the risk program only when re-anchored by the risk cap, at which point the coincidence itself certifies that the risk minima already sit on the cost–risk frontier. The American structure's premium advantage carries a mortality that this paper measures rather than assumes: re-simulated from the stress state itself with jump-adapted exact first-passage monitoring on 200,000 paths under the genuine asymmetric up/down jump calibration (statistically indistinguishable from the symmetrized-pool re-derivation, so the simplification the pricing engines rely on is not driving the result), the knock-out probability is 0.8925 (roughly eight times the level at which an all-KO book's survival-adjusted ledger can fit any allocation inside the budget), so the cheap hedge's protection is, with 89% probability, absent in precisely the scenario it is bought against. And the mixed vanilla/knock-out program shows this is a solvable allocation problem rather than a footnote: priced with an independently-derived standalone KO premium rather than the jointly-priced structure's Shapley attribution, feasible at every mortality level, switching instruments bang-bang at a closed-form indifference point of 6.16%, it walks the book to all-vanilla at the measured mortality and delivers lower residual volatility than the pure knock-out optimum under a strictly harder constraint. The static allocation question that precedes every dynamic hedging choice turns out, for this importer, to have an answer that is asymmetric, budget-pinned, instrument-aware, and – once the geometry is drawn – inevitable.

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